

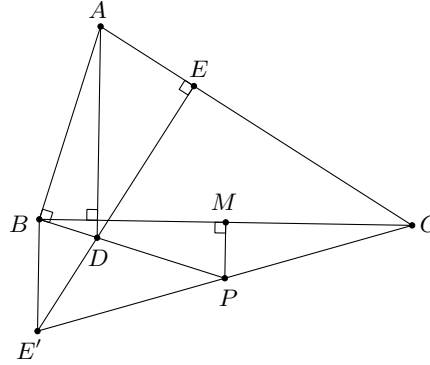


HK IMO Training 2019-2020
Problem Set 2 Suggested Solutions



Problem 5. Let ABC be a triangle. Let ℓ be the line through B perpendicular to AB . The perpendicular from A to BC meets ℓ at the point D . The perpendicular bisector of BC meets ℓ at the point P . Let E be the foot of the perpendicular from D to AC . Prove that triangle BPE is isosceles.

Solution 1. Let M be the midpoint of BC and E' the intersection of CP and DE . Let $\alpha = \angle DAB$.



Looking at the right-angled triangle we have $\angle BDA = 90^\circ - \alpha$ and so $\angle CBP = \angle CBD = \alpha$. Using symmetry in MP , $\angle E'CB = \angle PCB = \alpha$.

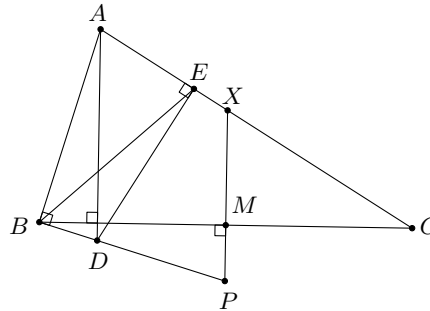
Now quadrilateral $ABDE$ is cyclic as $\angle ABD = 90^\circ = \angle DEA$. Hence $\angle E'EB = \angle DEB = \angle DAB = \alpha$.

Now, as $\angle E'CB = \alpha = \angle E'EB$, we have that quadrilateral $BE'CE$ is cyclic. As $\angle CEE' = 90^\circ$, the centre of this circle lies on CE' . But the centre also lies on the perpendicular bisector of BC so is P . In particular $PB = PE$. $\square$

Solution 2. First note that $\angle BDA = 90^\circ - \angle DAB = \angle ABC$. Also $ABDE$ is cyclic as $\angle ABD = 90^\circ = \angle DEA$. Thus $\angle BEA = \angle BDA = \angle ABC$. Therefore $\triangle ABC \sim \triangle AEB$ (they have two equal angles). In particular $\angle ABE = \angle BCA$ and so AB is tangent to the circle passing through B, C and E .

Now ℓ is perpendicular to AB so the centre of circle BCE lies on ℓ . But the centre of BCE must also lie on the perpendicular bisector of BC so is P . Therefore $PB = PE$. $\square$

Solution 3. Let M be the midpoint of BC and X the intersection of AC with MP .



Quadrilateral $ABDE$ is cyclic as $\angle ABD = 90^\circ = \angle DEA$. Also lines AD and XP are perpendicular to BC and so are parallel to each other. Thus $\angle CXP = \angle CAD = \angle EAD = \angle EBD = \angle EBP$ and so B, P, X, E are concyclic. In particular $\angle PEB = \angle PXB$. But, by symmetry in the perpendicular bisector XP , $\angle PXB = \angle CXP$. However we already know that $\angle CXP = \angle EBP$ so $\angle EBP = \angle PEB$, i.e. $\triangle BPE$ is isosceles with apex at P . Therefore $PB = PE$. $\square$

Problem 6. (a) Given are 2019 pairwise distinct integers having no odd prime divisors smaller than 37. Prove that there are two of these integers whose sum has no odd prime divisors smaller than 37 either.

(b) Does the statement remain correct if both occurrences of number 37 are replaced with 38?

Solution. (a) We assign to each number x the sequence

$$\bar{x} = (x_3, x_5, x_7, x_{11}, x_{13}, x_{17}, x_{19}, x_{23}, x_{29}, x_{31}),$$

where $x_p = 1$ if x leaves one of the remainders $1, 2, \dots, \frac{p-1}{2}$ upon division by p , and $x_p = -1$ otherwise. It is clear that if $x_p = y_p$, then $x + y$ is not a multiple of p . Since there are only $2^{10} = 1024$ possibilities for the sequence $\bar{x}$, there are given numbers x and y with $x \neq y$ and $\bar{x} = \bar{y}$, and then $x + y$ is not divisible by any of the primes $3, 5, \dots, 31$.

(b) In this case the assigned sequences are of length 11: $(x_3, x_5, \dots, x_{37})$. There are $2^{11} = 2048$ possible sequences, which we denote by $e_1, e_2, \dots, e_{2048}$. By the Chinese remainder theorem, for each $e_n = (x_3, x_5, \dots, x_{37})$ there is an integer y_n such that $y_n \equiv x_n \pmod{p}$ for $p = 3, 5, \dots, 37$. Every two assigned sequences e_i and e_j , where $1 \leq i < j \leq 2048$, differ at some position, say the one corresponding to a prime p , and then we have $p \mid y_i + y_j$. Hence the statement is now false. $\square$

Problem 7. In the country of Oddland, there are stamps with values 1 cent, 3 cent, 5 cent, etc, one type for each odd number. The rules of Odd land Postal Services stipulate the following: for any two distinct values, the number of stamps of the higher value on an envelope must never exceed the number of stamps of the lower value.

In the country of Squareland, on the other hand, there are stamps with values 1 cent, 4 cent, 9 cent, etc, one type for each square number. Stamps can be combined in all possible ways in Squareland without additional rules.

Prove for every positive integer n : In Oddland and Squareland there are equally many ways to correctly place stamps of a total value of n cent on an envelope. Rearranging the stamps on an envelope makes no difference.

Solution. We construct a bijection between possible combinations in Oddland and possible combinations in Squareland. Suppose we have a combination of Squareland stamps that sum to n cent, consisting of a_1 stamps of value 1 cent, a_2 stamps of value 4 cent, $\dots$, a_M stamps of value M^2 cent, so that

$$n = \sum_{k=1}^M k^2 a_k.$$

Now we express k^2 as $\sum_{j=1}^k (2j-1)$ and interchange the order of summation, which yields

$$n = \sum_{k=1}^M \sum_{j=1}^k (2j-1) a_k = \sum_{j=1}^M (2j-1) \sum_{k=j}^M a_k.$$

This gives us a possible combination of Oddland stamps: By setting $b_j = \sum_{k=j}^M a_k$, we have

$$n = \sum_{j=1}^M (2j-1) b_j.$$

This can be interpreted as a collection of b_1 stamps of value 1 cent, b_2 stamps of value 3 cent, $\dots$, b_M stamps of value $(2M-1)$ cent. We have $b_1 \geq b_2 \geq \dots \geq b_M$ by definition, so this is a legal combination in Oddland.

Conversely, if a combination in Oddland is given by the values $b_1, b_2, \dots, b_M$, we can use the identities $a_1 = b_1 - b_2, a_2 = b_2 - b_3, \dots, a_{M-1} = b_{M-1} - b_M, a_M = b_M$ to recover the corresponding combination in Squareland. (Note that these values are nonnegative whenever $b_1 \geq b_2 \geq \dots \geq b_M$.)

Since these two operations obviously are inverse to one another, we have found a bijection, which proves the statement. $\square$

Problem 8. Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfying the equality

$$f(f(x) + f(y)) = (x+y)f(x+y) \quad \text{for all } x, y \in \mathbb{R}. \quad (*)$$

Solution. Consider the following preliminary substitutions to (*):

- (1) $x = y = 0 \implies f(2f(0)) = 0$;
- (2) $x = 2f(0), y = 0 \implies f(f(0)) = 0$;
- (3) $x = y = f(0) \implies f(0) = 0$;
- (4) $y = 0 \implies f(f(x)) = xf(x)$ for all $x \in \mathbb{R}$.

Suppose that for some $a, b \in \mathbb{R}$, $f(a) = f(b) \neq 0$, then by (4),

$$af(a) = f(f(a)) = f(f(b)) = bf(b) \implies a = b.$$

We will show that there exists a $c \neq 0$ such that $f(c) = 0$.

Put $y = 1 - x$ into (*) to obtain

$$f(f(x) + f(1-x)) = f(1).$$

If $f(1) = 0$, then $c = 1$. Otherwise,

$$f(x) + f(1-x) = 1 \quad \text{for all } x \in \mathbb{R}. \quad (**)$$

Put $x = 0$ and $x = 1/2$ into (**) to obtain $f(1) = 1$ and $f(1/2) = 1/2$. Put $x = 1$ and $y = 1/2$ into (*) to obtain $f(3/2) = 3/2f(3/2)$, and so $f(3/2) = 0$.

Let c be a non-zero real number such that $f(c) = 0$. Put $x = y = c$ into (*) to obtain $f(2c) = 0$. Suppose that there exist a real number t such that $f(t) \neq 0$. Put $x = c, y = -c + t$ and $x = 2c, y = -2c + t$ successively to obtain

$$f(f(-c+t)) = tf(t) = f(f(-2c+t)),$$

where $tf(t) \neq 0$. Therefore, $f(-c+t) = f(-2c+t) \neq 0$, and so $-c+t = -2c+t$. This contradicts the assumption that $c \neq 0$. Consequently such a t does not exist and $f(x) = 0$ for all $x \in \mathbb{R}$.

Clearly, this function satisfies (*). $\square$